Database – No SQL – Mongodb

Session 9 –

1. **Install MongoDB Compass on your system, connect to your local MongoDB server, and create a new database named spotifyDB with a collection called playlists.**

Ans:

Open Compass and connect to your local MongoDB server using:

mongodb://localhost:27017

Create the database:

1. Click **Create Database**.
2. Database Name: spotifyDB
3. Collection Name: playlists
4. Click **Create Database**.

You should now have:

spotifyDB

└── playlists

2. **Using MongoDB Compass, import a JSON file containing at least 10 sample restaurant documents (name, cuisine, rating) into a new collection named zomatoRestaurants in your spotifyDB database.

Hint: You can find sample restaurant JSON data online or create your own small dataset.**

Ans:

Create a JSON file called restaurants.json with at least 10 documents:

```
[
  {
    "name": "Spice Garden",
    "cuisine": "Indian",
    "rating": 4.5
  },
  {
    "name": "Pasta Paradise",
    "cuisine": "Italian",
    "rating": 4.6
```

```
  },
  {
    "name": "Dragon Palace",
    "cuisine": "Chinese",
    "rating": 4.3
  },
  {
    "name": "Pizza Corner",
    "cuisine": "Italian",
    "rating": 4.4
  },
  {
    "name": "Curry House",
    "cuisine": "Indian",
    "rating": 4.2
  },
  {
    "name": "Sushi World",
    "cuisine": "Japanese",
    "rating": 4.7
  },
  {
    "name": "Burger Hub",
    "cuisine": "American",
    "rating": 4.1
  },
  {
    "name": "Tandoori Nights",
    "cuisine": "Indian",
    "rating": 4.6
  },
  {
    "name": "Mexican Fiesta",
    "cuisine": "Mexican",
    "rating": 4.3
  },
  {
    "name": "Cafe Mocha",
    "cuisine": "Cafe",
    "rating": 4.0
  }
]
```

3. **Sign up for a free MongoDB Atlas account, create a new cluster, and deploy a database named instaClone on Atlas.**

Ans:

Create a free MongoDB Atlas account:

[MongoDB Atlas]

Then:

1. Log in to Atlas.
2. Create a new project.
3. Click Create / Build a Database.
4. Select the available free-tier option if offered.
5. Give the cluster a name, such as instaCloneCluster.
6. Create a database user with a username and strong password.
7. Add your current IP address under Network Access.
8. Create the cluster.

Atlas creates databases when data is first inserted. You can create the target database/collection through Atlas Data Explorer:

Database: instaClone

Collection: zomatoRestaurants

4. **Export the zomatoRestaurants collection from your local MongoDB using the mongoexport tool and then import it into your instaClone database on MongoDB Atlas using the mongoimport tool.

Hint: Use the connection string provided by MongoDB Atlas for the import step.**

Ans:

Open a terminal where mongoexport is available:

```
mongoexport --uri="mongodb://localhost:27017/spotifyDB" \
  --collection=zomatoRestaurants \
  --out=restaurants_export.json \
  --jsonArray
```

This creates:

restaurants_export.json

containing the local zomatoRestaurants documents.

Import into MongoDB Atlas

In Atlas, click **Connect → Drivers** and copy your Atlas connection string.

It will look approximately like:

mongodb+srv://<username>:<password>@<cluster-url>/

Then run:

```
mongoimport --uri="mongodb+srv://<username>:<password>@<cluster-
url>/instaClone" \
  --collection=zomatoRestaurants \
  --file=restaurants_export.json \
  --jsonArray
```

5. **Connect to your MongoDB Atlas instaClone database using MongoDB Compass, and verify that all restaurant documents have been successfully imported by viewing the zomatoRestaurants collection and counting the documents.**

Ans:

Navigate to:

instaClone

└── zomatoRestaurants

You can also count them in the Compass collection interface. You should have:

10 documents

You can independently verify the count using mongosh:

mongosh "mongodb+srv://<username>:<password>@<cluster-url>/instaClone"

Then:

db.zomatoRestaurants.countDocuments()

Expected result:

10